

2019 1st tutorial

1. (a) 404 Note Alpha- Page 1 and 4

(b) 404 Note Alpha- Page 6

2. 404 Note Alpha - Page 30

3. (a) 404 Note Alpha- Page 11

(b) 404 Note Alpha- Page 14

4. a) **The role of medicine in the historical decline of the mortality rate is small, perhaps negligible. Explain this statement with empirical evidence.**

Chatgpt: There is a growing body of evidence that suggests that medicine has played a relatively small role in the historical decline of mortality rates. While medical interventions have certainly contributed to improvements in health outcomes, the impact of medicine is often overstated when it comes to reducing mortality rates.

One of the most compelling pieces of evidence for this claim comes from historical trends in mortality rates. For example, during the 20th century, mortality rates declined rapidly in many countries around the world. However, most of these declines occurred before the widespread availability of many of the medical interventions that we now take for granted. In fact, some of the most significant declines in mortality occurred before the introduction of antibiotics, vaccines, and other modern medical technologies.

Moreover, the decline in mortality rates has been shown to be associated with a variety of social and economic factors, such as improvements in sanitation, nutrition, and living conditions. For example, improvements in sanitation have been linked to declines in infectious diseases like cholera and typhoid fever. Similarly, improvements in nutrition have been associated with declines in conditions like malnutrition and rickets.

While medical interventions have certainly played a role in reducing mortality rates, the evidence suggests that their contribution has been relatively small compared to other factors. This is not to say that medicine is unimportant, but rather that we need to look beyond medical interventions to fully understand the historical decline in mortality rates.

- b) **According to Victor Fuchs what factors are responsible for the historical decline in infant mortality rate. explain. Does this mean medical research is unimportant? explain.**

Chatgpt: Victor Fuchs, a prominent health economist, has identified a range of factors that have contributed to the historical decline in infant mortality rates. These include improvements in nutrition,

sanitation, housing, public health measures, and medical interventions. However, Fuchs emphasizes that the relative importance of each of these factors has changed over time, and that improvements in non-medical factors have played a particularly significant role in recent decades.

In the early part of the 20th century, medical interventions such as immunizations and antibiotics played a key role in reducing infant mortality rates. However, as these interventions became more widely available and effective, the relative importance of non-medical factors increased. For example, improvements in sanitation and public health measures such as access to clean water and better waste management have had a significant impact on reducing the incidence of infectious diseases that can cause infant mortality.

Similarly, improvements in nutrition have been a critical factor in reducing infant mortality rates. Better maternal nutrition during pregnancy and improved infant feeding practices have led to significant improvements in infant health outcomes. Additionally, advances in housing conditions have played a role, with improved living standards reducing exposure to environmental hazards such as lead and other toxins.

It is important to note that the role of medical research and interventions in improving health outcomes cannot be discounted. Many medical interventions, such as vaccinations and antibiotics, have been essential in reducing the incidence of infectious diseases and improving overall health outcomes. However, as Fuchs suggests, a holistic understanding of the factors that contribute to health outcomes is essential, and non-medical factors such as improved sanitation, nutrition, and housing conditions have played a significant role in the historical decline in infant mortality rates.

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1. A) 404 Note Alpha- Page 4
b) 404 Note Alpha- Page 6

2.a) **If not medicine what factors were responsible for the historical decline in the population mortality rate? discuss.**

Chatgpt: The historical decline in population mortality rates can be attributed to a range of factors, some of which are related to medical interventions, but many of which are not. Some of the most significant factors that have contributed to the decline in mortality rates include:

Improved public health measures: In the early part of the 20th century, public health measures such as improved sanitation, clean water supply, and waste management played a crucial role in reducing mortality rates from infectious diseases like cholera and tuberculosis.

Better nutrition: Improved access to food and better nutritional practices have led to significant improvements in health outcomes, particularly for children. The development and distribution of nutritional supplements, fortified foods, and improved infant feeding practices have helped to reduce mortality rates from malnutrition and other nutrition-related diseases.

Safer working conditions: Improvements in occupational health and safety practices, particularly in industrialized countries, have reduced mortality rates from work-related injuries and illnesses.

Improved housing conditions: Access to safe and healthy housing has played a critical role in reducing mortality rates. Better living conditions have reduced exposure to environmental hazards such as lead, asbestos, and other toxins.

Education and public awareness: Improvements in education and public awareness campaigns have led to better health outcomes by promoting healthy behaviors and reducing risky behaviors such as smoking.

Medical interventions: While the impact of medical interventions on reducing mortality rates may be smaller than commonly thought, medical advances have played a critical role in improving health outcomes. Vaccinations, antibiotics, and other medical interventions have reduced mortality rates from infectious diseases and improved survival rates for a range of medical conditions.

In conclusion, while medical interventions have played a critical role in reducing mortality rates, improvements in public health measures, nutrition, working conditions, housing conditions, education, and public awareness have also been significant factors in the historical decline in population mortality rates. A holistic approach that considers the interplay of these various factors is necessary to continue to improve health outcomes in the future.

b) 404 Note Alpha- Page 14

3.a) 404 Note Alpha- Page 11

b) answer in previous 4b chatgpt

4. A) 404 Note Alpha - Page 22

b) 404 Note Alpha- Page 16

c) 404 Note Alpha- Page 23

5.a) Explain, in the light of grossman model how an individual chooses optimum amount of income and leisure.

The Grossman model, developed by economist Michael Grossman, is an economic model that explores how individuals make decisions regarding their allocation of time between work and leisure. The model considers both the economic benefits derived from working and the utility gained from leisure activities. In this model, individuals aim to maximize their overall well-being or utility by choosing the optimal combination of income and leisure time.

The basic premise of the Grossman model is that individuals face a trade-off between income and leisure. When individuals work, they allocate time and effort to income-generating activities, which

increases their income but reduces their available leisure time. Conversely, when individuals choose more leisure time, they sacrifice potential income. To determine the optimal allocation of income and leisure, individuals consider two main factors: the utility gained from leisure and the utility gained from consumption (income). The utility gained from leisure represents the satisfaction or enjoyment derived from non-work activities, such as spending time with family, pursuing hobbies, or engaging in recreational activities. The utility gained from consumption represents the satisfaction derived from the goods and services individuals can purchase with their income.

The Grossman model assumes that individuals have a preference for both leisure and consumption, and these preferences can be quantified through utility functions. These utility functions reflect the individual's preferences and can vary across individuals. The model assumes that individuals seek to maximize their overall utility, subject to their budget constraint, which relates to the trade-off between income and leisure.

The budget constraint is determined by the wage rate and the number of hours an individual can work. When an individual chooses to work, they receive income proportional to the wage rate multiplied by the number of hours worked. The individual's total available time is limited, and the time spent working reduces the time available for leisure. As a result, the individual must decide how many hours to allocate to work and how many hours to allocate to leisure, considering the associated benefits and costs.

The optimum allocation of income and leisure occurs when the marginal utility gained from an additional hour of leisure equals the marginal utility gained from an additional unit of income. The marginal utility represents the additional satisfaction gained from a small change in the quantity of leisure or income. By equating the marginal utilities, individuals achieve a balanced allocation that maximizes their overall utility.

However, it's important to note that the Grossman model is a simplification of reality and assumes certain assumptions and constraints. It doesn't consider factors such as non-monetary benefits of work, variations in individual preferences and constraints, or the presence of market imperfections. Nonetheless, the model provides a framework for understanding how individuals make decisions about income and leisure, based on the trade-offs they face and their preferences for utility.

b) How is it possible to earn more money and leisure time together? Explain in the light of grossman model. Also in 404 Note Alpha- Page 20

In the Grossman model, individuals face a trade-off between income and leisure, suggesting that it is not possible to simultaneously increase both income and leisure time. The model assumes that as individuals allocate more time to work to earn higher income, their available leisure time decreases.

However, it is worth noting that the Grossman model focuses on the individual's decision-making process and assumes a certain set of constraints and trade-offs. While the model suggests that individuals face a trade-off between income and leisure, it doesn't capture all possible scenarios and factors that could influence the outcome.

In reality, there can be situations or strategies that allow individuals to increase both income and leisure time to some extent. Here are a few possibilities to consider:

1. Increased productivity: If an individual becomes more productive in their work, they may be able to generate higher income within a given amount of time. By becoming more efficient or acquiring new skills, they can achieve higher income without significantly increasing their working hours. This increased productivity can create opportunities to earn more money while maintaining or even reducing working hours, thus leaving more time for leisure.

2. Flexibility and autonomy: Some individuals may have the ability to negotiate flexible work arrangements or work on a freelance or self-employed basis. This flexibility can provide more control over one's schedule, allowing individuals to allocate time according to their preferences and potentially increase leisure time while still earning a satisfactory income.

3. Passive income and investments: Investing in assets that generate passive income, such as rental properties, stocks, or businesses, can provide a means of earning income without active involvement or significant time commitment. By generating income from such investments, individuals may have the opportunity to reduce their working hours and enjoy more leisure time while maintaining or increasing their overall income.

4. Work-life balance and prioritization: Individuals can strive for a better work-life balance by consciously allocating time to activities that are personally fulfilling and provide relaxation and rejuvenation. By setting boundaries and priorities, individuals can optimize their leisure time while still focusing on their careers and earning income.

It's important to note that these possibilities may not always be feasible or applicable to every individual, as they depend on personal circumstances, skills, opportunities, and external factors. The Grossman model provides a theoretical framework to understand the general trade-off between income and leisure, but real-world situations can deviate from this simplified model due to various complexities and individual differences.

2nd and 3rd tutorial without year

1. Math from 404 Note Beta Need a clear version
2. A} Basics from 404 Note Beta
B) 404 Note Beta page 23
c) discuss the determinants of market health care visit functions.

Determinants of market healthcare visit functions refer to the factors that influence individuals' decisions to seek medical care and the demand for healthcare services in the market. Several key determinants can shape healthcare visit functions:

1. **Price and Cost:** The cost of healthcare services, including consultation fees, diagnostic tests, medications, and insurance coverage, plays a crucial role in determining healthcare visit functions. Higher prices or out-of-pocket costs may deter individuals from seeking medical care, especially for non-emergency or preventive services.
2. **Income and Affordability:** The level of individuals' income and their ability to afford healthcare services significantly impact healthcare visit functions. Lower-income individuals may be more sensitive to costs and have a higher likelihood of forgoing or delaying healthcare visits due to financial constraints.
3. **Health Insurance Coverage:** The presence and extent of health insurance coverage influence healthcare visit functions. Insured individuals may be more likely to seek medical care as they have reduced financial barriers. The type of insurance (e.g., private, employer-provided, government-funded) and the extent of coverage (e.g., deductibles, copayments) can also affect individuals' decisions.
4. **Perceived Need and Health Status:** Individuals' perception of their health needs and their subjective assessment of their health status play a vital role in determining healthcare visit functions. The severity of symptoms, perceived urgency, and concerns about one's health condition can motivate individuals to seek medical care.
5. **Demographics:** Certain demographic factors can influence healthcare visit functions. For example, age, gender, and family composition may affect healthcare utilization patterns. Children, elderly individuals, and women of reproductive age often have specific healthcare needs and may have higher healthcare visit rates.
6. **Health Knowledge and Awareness:** Individuals' knowledge about health issues, preventive care, and the benefits of early intervention can influence healthcare visit functions. Greater health literacy and awareness may lead individuals to seek medical care more proactively.
7. **Accessibility and Availability:** The availability and accessibility of healthcare services, including the proximity of healthcare facilities, appointment wait times, and the availability of healthcare professionals, can impact healthcare visit functions. Inadequate access to healthcare services may hinder individuals from seeking necessary care.

8. Cultural and Social Factors: Cultural beliefs, social norms, and personal preferences can shape healthcare visit functions. Cultural factors may influence individuals' beliefs about illness, treatment options, and the role of healthcare providers. Social support systems and peer influence can also affect healthcare-seeking behavior.

9. Government Policies and Regulations: Government policies, such as healthcare regulations, subsidies, and incentives, can impact healthcare visit functions. Policy interventions like universal healthcare coverage or targeted programs can influence individuals' decisions to seek medical care.

It's important to note that these determinants interact and vary across individuals, communities, and healthcare systems. The complex interplay of these factors influences the demand for healthcare services and shapes the overall market healthcare visit functions.

4) 404 Insurance Math

5) Shikkhok Consultancy Needed

6) a) 404 Note Beta page 1-5 basics

b) Shikkhok Consultancy

7) 404- Asymmetric Information.pdf all of it

8) 404 Getis from 30 page onwards

Final 2018

1) Already Covered

2) A) 404 Getis page 20

b) already covered

3) A) 404 Note Alpha page 23

b) 404 Note Alpha page 26

c) limitations of Grossman Model

While the Grossman model provides valuable insights into the decision-making process regarding income and leisure allocation, it also has some limitations. Here are a few key limitations of the Grossman model:

1. Simplified Assumptions: The Grossman model makes several simplifying assumptions to facilitate analysis, such as assuming individuals have perfect information, make rational decisions, and have homogeneous preferences. In reality, individuals may have limited

information, exhibit bounded rationality, and have diverse preferences, which can significantly impact their choices regarding income and leisure.

2. Static Framework: The Grossman model is primarily a static model that examines the trade-off between income and leisure at a given point in time. It doesn't account for dynamic factors, such as changing preferences, evolving labor market conditions, technological advancements, or life-cycle considerations. These dynamic factors can influence an individual's decisions and alter their optimal allocation of income and leisure over time.

3. Neglecting Non-Monetary Factors: The model focuses primarily on the monetary benefits derived from income and leisure choices and assumes that individuals are solely motivated by monetary utility. However, individuals also derive non-monetary benefits from work, such as social interactions, skill development, or personal fulfillment. Neglecting these non-monetary factors may limit the model's applicability in capturing the complexities of individuals' decision-making processes.

4. Ignoring Market Imperfections: The Grossman model assumes perfect competition and efficient markets, disregarding market imperfections, such as wage rigidities, information asymmetry, or barriers to entry. In reality, these imperfections can affect an individual's income-earning potential, limit job choices, and influence the optimal allocation of income and leisure.

5. External Influences: The model doesn't explicitly consider external influences, such as social norms, cultural factors, or institutional constraints, which can shape individuals' decisions regarding income and leisure. These external influences can have a significant impact on an individual's choices and may deviate from the predictions of the model.

6. Heterogeneity and Individual Differences: The Grossman model treats individuals as homogeneous entities with the same preferences and constraints. However, individuals have diverse characteristics, values, and constraints that can lead to different decisions regarding income and leisure. The model's inability to account for heterogeneity limits its ability to capture the full range of individual behaviors and choices.

Despite these limitations, the Grossman model serves as a valuable theoretical framework for understanding the basic trade-offs individuals face between income and leisure. By recognizing these limitations and incorporating additional complexities, researchers can build upon the model to develop more realistic and comprehensive frameworks for analyzing individual decision-making in the context of income and leisure allocation.

- 4) 404 Note Beta page 31 math
- 5) Already covered
- 6) Asymmetric Sheet
- 7) Shikhhok'

- 8) Most likely books covering the topic SID

FIRST tutorial 2017

1,2,3,4,5,6 already covered up the whole questions (HAIL SHOWKOT)

2nd tutorial shal nai

Part A- 1,2,3 covered.

Part B- 4,5,6 (A) covered

6 (b) shikhhok

3rd tutorial 2017

1. Asymmetric sheet
2. A) book or chatgpt
b) Maths from our 3rd tutorial picture provided
3. what is discounting? why do we discount future value into its present value? **Also in CBA Sheet**

Discounting refers to the process of adjusting the value of future cash flows or benefits to reflect their present value. It involves calculating the present value of a future amount by considering the time value of money. The underlying principle is that a dollar received in the future is worth less than a dollar received today due to the opportunity cost of not having that money available for use or investment in the meantime.

We discount future value into its present value for several reasons:

1. **Time Value of Money:** The concept of the time value of money recognizes that the value of money changes over time. Money has potential earning capacity through investment or interest, and individuals generally prefer to have money available to them in the present rather than in the future. By discounting future value, we account for this preference and reflect the reduced value of money over time.

2. Risk and Uncertainty: Future cash flows are uncertain and carry a degree of risk. By discounting future cash flows to their present value, we consider the risk associated with receiving the money in the future. Uncertainty and the potential for risks to materialize reduce the value of future cash flows, and discounting helps incorporate this risk into present value calculations.

3. Opportunity Cost: By discounting future cash flows, we acknowledge the opportunity cost of not having access to that money in the present. If the funds were available today, they could be used for various purposes, such as investment, consumption, or debt repayment. By discounting, we recognize that by committing funds to a future cash flow, we forego alternative uses of that money in the present.

4. Consistency and Comparability: Discounting future cash flows allows for consistent and comparable analysis of different financial streams. By converting future cash flows into present value, we can compare and evaluate various investment options or financial decisions on an equal footing. It enables decision-makers to assess the value of different choices at a specific point in time.

The process of discounting is typically performed using a discount rate, which represents the required rate of return or the cost of capital. The discount rate reflects the risk and opportunity cost associated with the cash flows. Different methodologies, such as net present value (NPV) and discounted cash flow (DCF) analysis, employ discounting to determine the present value of future cash flows and aid decision-making.

In summary, discounting future value into its present value is essential because it incorporates the time value of money, considers the risk and uncertainty associated with future cash flows, acknowledges opportunity costs, and allows for consistent and comparable analysis of financial decisions.

b) how do we value human life? explain. Also in CBA Sheet

Valuing human life is a complex and ethically challenging task, and different approaches have been developed to address this issue. It is important to note that assigning a monetary value to human life is a subject of debate and varies across different contexts and perspectives. Here are a few common approaches used to value human life:

1. Human Capital Approach: The human capital approach is an economic method that values human life based on the potential earnings and productivity of an individual over their lifetime. It considers factors such as education, skills, work experience, and expected future income. This approach estimates the value of a human life by quantifying the economic contributions that would be lost if an individual were to die prematurely.

2. Willingness-to-Pay (WTP) Approach: The willingness-to-pay approach attempts to estimate the value of human life based on individuals' preferences and the amount they are willing to pay to reduce the risk of death or injury. It often involves analyzing data from surveys or behavioral experiments that assess people's willingness to trade money for reductions in mortality risks. This approach derives the value of human life indirectly from individuals' revealed preferences.

3. Quality-Adjusted Life Years (QALYs): The concept of quality-adjusted life years is often used in healthcare and health economics to assess the value of medical interventions or health policies. QALYs combine both the quantity and quality of life, considering the impact of health conditions on an individual's well-being. By assigning weights to different health states, including life-threatening conditions or disabilities, this approach allows for comparisons of different health outcomes in terms of their value.

4. Cost of Statistical Life (VSL): The cost of statistical life is an approach used by governments and regulatory agencies to make policy decisions related to safety regulations or cost-benefit analyses. It estimates the value of a statistical life by analyzing data on compensation for workplace fatalities or using hedonic wage studies that measure the trade-off between wages and occupational risks. This approach considers the value society places on reducing risks to human life.

5. Ethical and Humanistic Approaches: Some argue that human life should not be assigned a monetary value due to its inherent worth and dignity. Ethical and humanistic perspectives prioritize the intrinsic value of human life and argue that attempting to quantify it with monetary terms is inappropriate. These approaches emphasize the importance of human rights, fairness, and social justice in decision-making processes.

It is important to recognize that these approaches have their limitations and critics. They often oversimplify the complex and multifaceted nature of human life, ignore subjective factors, and can lead to ethical dilemmas. Furthermore, assigning a monetary value to human life does not imply that it is acceptable to trade or disregard human lives solely based on economic considerations.

Valuing human life is a topic that involves moral, ethical, and societal considerations, and the approaches mentioned above are attempts to provide frameworks for decision-making in certain contexts. Ultimately, the value of human life is deeply subjective and influenced by cultural, social, and individual perspectives.

4.explain the basic principles of cba? Also in CBA Sheet

Final 2017

1,2,3,4 covered

5. 404 Note Beta page 52 math

6,7,8 covered

Final 2016

1 Covered

2 A) use chatgpt for definitions

b) explain the basic principles of cost benefit analysis. CBA Sheet eo ache.

Cost-benefit analysis (CBA) is a systematic approach used to assess the economic feasibility of a project, policy, or decision by comparing the costs incurred with the benefits gained. The fundamental principles of cost-benefit analysis include:

1. Identifying and Quantifying Costs and Benefits: The first step in cost-benefit analysis is to identify and quantify the costs and benefits associated with the decision under consideration. Costs include both direct expenditures (such as investment costs, operating costs, and maintenance costs) and opportunity costs (the value of resources used in the project that could have been used elsewhere). Benefits encompass both monetary and non-monetary impacts and can include increased revenues, cost savings, improved quality of life, environmental gains, or other desirable outcomes.

2. Monetary Valuation: In cost-benefit analysis, costs and benefits need to be expressed in a common unit of measurement, usually monetary terms. This allows for comparability and aggregation of different costs and benefits. Monetary valuation involves assigning a monetary value to each cost and benefit, which can be challenging for non-market goods or intangible impacts. Various valuation techniques and methods, such as market prices, surveys, or stated preference methods, may be used to estimate monetary values.

3. Time Discounting: Time discounting is the process of adjusting the value of future costs and benefits to their present value. Since costs and benefits occur over time, they are not considered equally valuable in the present. Future costs and benefits are discounted to reflect the time preference for consumption and the opportunity cost of capital. The discount rate is typically applied to adjust future values to their equivalent present values.

4. Comparison and Net Present Value: In cost-benefit analysis, the net present value (NPV) is calculated by subtracting the discounted costs from the discounted benefits. The NPV represents the net economic value of the project or decision. If the NPV is positive, it indicates that the benefits outweigh the costs, suggesting that the project or decision is economically viable. Conversely, a negative NPV implies that the costs exceed the benefits, signaling potential economic inefficiency.

5. Sensitivity Analysis: Sensitivity analysis is performed to assess the impact of variations in key inputs or assumptions on the results of the cost-benefit analysis. By exploring different scenarios and ranges of values, decision-makers can understand the robustness and uncertainties associated with the estimated costs and benefits. Sensitivity analysis helps identify critical factors that influence the outcome and assists in decision-making under uncertainty.

6. Distributional Analysis: In addition to the overall net benefits, cost-benefit analysis can consider the distribution of costs and benefits across different stakeholders or groups in society. This allows for an examination of potential equity or fairness implications and helps policymakers make informed decisions considering different social perspectives.

By applying these basic principles, cost-benefit analysis provides a structured framework for assessing the economic efficiency and desirability of projects or policies. It aids decision-makers in comparing different alternatives and making informed choices by considering both the costs incurred and the benefits achieved.

3,4,5,6,7 covered

8 shikkhok

1st tutorial shal nai

1,2 covered

3. 404 Note Beta page 33 math
4. Covered
5. 31 page of 404 Note Beta somewhat similar math

Kisui Lekha nai Shawon Bissas

1,2 maths already similar covered in 404 Note Beta

3,4,5 chat gpt and textbook

Final 2015

1-8 already covered

47 First Tutorial

404 1st Tutorial Note.pdf each and one of them (Last provided by Faiaz in Group)

47 2nd Tutorial

1. A) not found
b) not found
c) not found

2. a) not found

b) not found

3.a) 404 Note Beta 13 page

b) not found

47 3rd tutorial

1. A) cba sheet page 1
b) math from picture
2. a) cba sheet page-4
b) math from picture

3. Uncertainty Sheet by Sir